Topic 5: Generators

Each problem is worth 2 points (one point per printed line).

Problem 1. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def g():
    yield 1
    yield 2
    yield 3
try:
    print('list(g())=', list(g()))
    print('sum(g())=', sum(g()))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 16 / 16 = 1.00

Problem 2. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def g():
    for i in [1, 2, 3]:
        i
try:
    print('g()=', g())
    print('list(g())=', list(g()))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 3. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def g():
    return
    yield 1
try:
    print('list(g())=', list(g()))
    print('sum(g())=', sum(g()))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 4. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def countdown(n):
    for i in range(n, 0, -1):
        yield i
try:
    print('list(countdown(5))=', list(countdown(5)))
    print('sum(countdown(5))=', sum(countdown(5)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 5. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def countdown(n):
    for i in range(n + 1, 0, -1):
        yield i
try:
    print('list(countdown(3))=', list(countdown(3)))
    print('sum(countdown(3))=', sum(countdown(3)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 6. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def g():
    yield 1
    yield 2
    yield 3
try:
    it = g()
    print('list(it)=', list(it))
    print('sum(it)=', sum(it))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 7. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def g():
    yield 1
    yield 2
    yield 3
try:
    it = g()
    print('next(it)=', next(it))
    print('list(it)=', list(it))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 8. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def countdown(n):
    for i in range(n, 0, -1):
        yield i
try:
    print('list(countdown(3))=', list(countdown(3)))
    print('next(countdown(0))=', next(countdown(0)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 9. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1
try:
    it = nats()
    print('next(it)=', next(it))
    print('next(it)=', next(it))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 10. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        i += 1
        yield i
try:
    it = nats()
    print('next(it)=', next(it))
    print('next(it)=', next(it))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 11. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def fib():
    a, b = 0, 1
    while True:
        yield a
        a, b = b, a + b
try:
    it = fib()
    xs = [next(it) for _ in range(6)]
    print('first3=', xs[:3])
    print('last3=', xs[3:])
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 12. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def fib():
    a, b = 0, 1
    while True:
        yield a
        a = b
        b = a + b
try:
    it = fib()
    xs = [next(it) for _ in range(6)]
    print('first3=', xs[:3])
    print('last3=', xs[3:])
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 13. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1
try:
    print('start')
    print('list=', list(nats()))
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 14. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [1, 2, 3, 4]
try:
    print('list(xx)=', list(x * x for x in xs))
    print('sum(xx)=', sum(x * x for x in xs))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 15. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [1, 2, 3, 4]
n = 2
gen = (n * x for x in xs)
n = 3
try:
    print('n now=', n)
    print('list(gen)=', list(gen))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 16. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [[1, 2], [3, 4], [5]]
try:
    print('list=', list(y for x in xs for y in x))
    print('sum=', sum(y for x in xs for y in x))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 17. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1
try:
    print('start')
    print('list=', list(x for x in nats()))
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 18. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1

def take(g, n):
    for _ in range(n):
        yield next(g)
try:
    print('list(t)=', list(take(nats(), 5)))
    print('sum(t)=', sum(take(nats(), 4)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 19. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1

def take(g, n):
    for _ in range(n - 1):
        yield next(g)
try:
    print('list(t)=', list(take(nats(), 5)))
    print('sum(t)=', sum(take(nats(), 4)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 20. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1

def take(g, n):
    for _ in range(n):
        yield next(g)
try:
    g = nats()
    print('take=', list(take(g, 3)))
    print('next(g)=', next(g))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 21. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1

def take(g, n):
    for _ in range(n):
        yield next(g)
        next(g)
try:
    print('list(t)=', list(take(nats(), 4)))
    print('sum(t)=', sum(take(nats(), 3)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 22. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def small():
    yield 'a'
    yield 'b'

def take(g, n):
    for _ in range(n):
        yield next(g)
try:
    print('take2=', list(take(small(), 2)))
    print('take3=', list(take(small(), 3)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 23. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [1, 2, 3, 4]
ys = ['a', 'b', 'c']
try:
    print('zip(xs, ys)=', list(zip(xs, ys)))
    print('zip(ys, xs)=', list(zip(ys, xs)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 24. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def g():
    yield 1
    yield 2
    yield 3
    yield 4
try:
    it1 = g()
    print('zip2=', list(zip(it1, it1)))
    it2 = g()
    print('zip3=', list(zip(it2, it2, it2)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 25. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def g():
    yield 'a'
    yield 'b'
    yield 'c'
try:
    it = g()
    print('enum=', list(enumerate(it)))
    print('again=', list(enumerate(it)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 26. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [1, 2, 3, 4]
ys = [10, 20, 30]
try:
    print('map=', list(map(lambda a, b: a + b, xs, ys)))
    print('map2=', list(map(lambda a, b: a * b, xs, ys)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 27. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
xs = [0, 1, '', 'a', [], [1], None, False]
try:
    print('filter=', list(filter(None, xs)))
    print('count=', len(list(filter(None, xs))))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 28. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1
try:
    print('start')
    print('list=', list(map(lambda x: x * x, nats())))
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 29. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
try:
    print('islice=', list(itertools.islice(itertools.count(0, 2), 5)))
    print('sum=', sum(itertools.islice(itertools.count(1, 3), 4)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 30. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
try:
    print('islice=', list(itertools.islice(itertools.count(1, 2), 5)))
    print('sum=', sum(itertools.islice(itertools.count(0, 3), 4)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 31. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
try:
    print('cycle=', list(itertools.islice(itertools.cycle([1, 2, 3]), 7)))
    print('sum=', sum(itertools.islice(itertools.cycle([4, 5]), 5)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 32. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
try:
    it = itertools.chain([1, 2, 3], [4, 5, 6])
    print('first=', list(itertools.islice(it, 4)))
    print('rest=', list(it))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 33. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
try:
    print('start')
    print('list=', list(itertools.cycle([1, 2, 3])))
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 34. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
def fib():
    a, b = 0, 1
    while True:
        yield a
        a, b = b, a + b
try:
    print('sum=', sum(itertools.islice(map(lambda x: x * x, fib()), 5)))
    print('list=', list(itertools.islice(map(lambda x: 2 * x, fib()), 5)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 35. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
def fib():
    a, b = 0, 1
    while True:
        yield a
        a, b = b, a + b
try:
    print('sum=', sum(itertools.islice(map(lambda x: x * x, fib()), 4)))
    print('list=', list(itertools.islice(map(lambda x: 2 * x, fib()), 6)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 36. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
try:
    print('a=', list(itertools.islice(itertools.chain([1, 2, 3], itertools.count(0)), 6)))
    print('b=', sum(itertools.islice(itertools.chain([1, 2, 3], itertools.count(0)), 5)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 37. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
import itertools
def nats():
    i = 0
    while True:
        yield i
        i += 1
try:
    print('evens=', list(itertools.islice(filter(lambda x: x % 2 == 0, nats()), 5)))
    print('threes=', list(itertools.islice(filter(lambda x: x % 3 == 0, nats()), 5)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 38. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def take(g, n):
    for _ in range(n):
        yield next(g)
try:
    print('a=', list(take(map(lambda x: x * x, [1, 2, 3, 4]), 2)))
    print('b=', sum(take(map(lambda x: x + 1, [1, 2, 3, 4]), 3)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 39. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def nats():
    i = 0
    while True:
        yield i
        i += 1
try:
    print('start')
    print('list=', list(filter(lambda x: True, nats())))
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 40. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Countdown:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1
try:
    print('list=', list(Countdown(3)))
    print('sum=', sum(Countdown(3)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 41. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Countdown:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        self.i -= 1
        return self.i + 1
try:
    print('start')
    print('list=', list(Countdown(3)))
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 42. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Countdown:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i
try:
    print('list=', list(Countdown(3)))
    print('sum=', sum(Countdown(3)))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 43. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class CountdownIter:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1

class Countdown:
    def __init__(self, n):
        self.n = n
    def __iter__(self):
        return CountdownIter(self.n)
try:
    c = Countdown(3)
    print('a=', list(c), 'b=', list(c))
    print('c=', list(c), 'sum=', sum(c))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 44. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Countdown:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1
try:
    c = Countdown(3)
    print('first=', list(c))
    print('second=', list(c))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 45. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Countdown:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1
try:
    c = Countdown(3)
    it = iter(c)
    print('next=', next(it))
    print('rest=', list(it))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 46. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def countdown(n):
    for i in range(n, 0, -1):
        yield i

class CountdownIter:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1

class Countdown:
    def __init__(self, n):
        self.n = n
    def __iter__(self):
        return CountdownIter(self.n)
try:
    c1 = countdown(3)
    print('iter(c1) is c1:', iter(c1) is c1)
    c2 = Countdown(3)
    print('iter(c2) is c2:', iter(c2) is c2)
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 47. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class CountdownIter:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1

class Countdown:
    def __init__(self, n):
        self.n = n
    def __iter__(self):
        return CountdownIter(self.n)
try:
    c = Countdown(3)
    print('sum1=', sum(c))
    print('sum2=', sum(c))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 48. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Countdown:
    def __init__(self, n):
        self.i = n
    def __iter__(self):
        return self
    def __next__(self):
        if self.i <= 0:
            raise StopIteration
        self.i -= 1
        return self.i + 1
try:
    c = Countdown(3)
    print('for=', [x for x in c])
    print('next after for=', next(c))
except StopIteration:
    print('StopIteration')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 49. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Range:
    def __init__(self, start, stop, step=1):
        self.start = start
        self.stop = stop
        self.step = step
    def __iter__(self):
        i = self.start
        while i < self.stop:
            yield i
            i += self.step
try:
    print('list=', list(Range(0, 5)))
    print('sum=', sum(Range(0, 5)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 50. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Range:
    def __init__(self, start, stop, step=1):
        self.start = start
        self.stop = stop
        self.step = step
    def __iter__(self):
        i = self.start
        while i <= self.stop:
            yield i
            i += self.step
try:
    print('list=', list(Range(0, 5)))
    print('sum=', sum(Range(0, 5)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 51. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Range:
    def __init__(self, start, stop, step=1):
        self.start = start
        self.stop = stop
        self.step = step
    def __iter__(self):
        i = self.start
        while i < self.stop:
            yield i
            i += self.step + 1
try:
    print('list=', list(Range(0, 10, 3)))
    print('sum=', sum(Range(0, 10, 3)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 52. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Range:
    def __init__(self, start, stop, step=1):
        self.start = start
        self.stop = stop
        self.step = step
    def __iter__(self):
        i = self.start
        while i < self.stop:
            yield i
            i += self.step

class RangeGen:
    def __init__(self, start, stop, step=1):
        self.gen = (i for i in range(start, stop, step))
    def __iter__(self):
        return self.gen
try:
    r1 = Range(0, 4)
    print('r1 twice=', list(r1), list(r1))
    r2 = RangeGen(0, 4)
    print('r2 twice=', list(r2), list(r2))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 53. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
def countdown(n):
    for i in range(n, 0, -1):
        yield i

class Range:
    def __init__(self, start, stop):
        self.start = start
        self.stop = stop
    def __iter__(self):
        i = self.start
        while i < self.stop:
            yield i
            i += 1
try:
    g = countdown(5)
    print('g: sum=', sum(g), 'then list=', list(g))
    r = Range(0, 5)
    print('r: sum=', sum(r), 'then list=', list(r))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 54. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Range:
    def __init__(self, start, stop, step=1):
        self.start = start
        self.stop = stop
        self.step = step
    def __iter__(self):
        i = self.start
        while i < self.stop:
            return i
            i += self.step
try:
    print('start')
    print('list=', list(Range(0, 10, 0)))
except MemoryError:
    print('MemoryError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 55. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Node:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None

def insert(root, val):
    if root is None:
        return Node(val)
    if val < root.val:
        root.left = insert(root.left, val)
    elif val > root.val:
        root.right = insert(root.right, val)
    return root

def inorder(t):
    if t is None:
        return
    yield from inorder(t.left)
    yield t.val
    yield from inorder(t.right)

class Tree:
    def __init__(self):
        self.root = None
    def insert(self, val):
        self.root = insert(self.root, val)
    def __iter__(self):
        return inorder(self.root)
try:
    t = Tree()
    for v in [5, 3, 8, 1, 4]:
        t.insert(v)
    xs = list(t)
    print('first3=', xs[:3])
    print('last2=', xs[-2:])
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 56. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Node:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None

def insert(root, val):
    if root is None:
        return Node(val)
    if val < root.val:
        root.left = insert(root.left, val)
    elif val > root.val:
        root.right = insert(root.right, val)
    return root

def preorder(t):
    if t is None:
        return
    yield t.val
    yield from preorder(t.left)
    yield from preorder(t.right)

class Tree:
    def __init__(self):
        self.root = None
    def insert(self, val):
        self.root = insert(self.root, val)
    def __iter__(self):
        return preorder(self.root)
try:
    t = Tree()
    for v in [5, 3, 8, 1, 4]:
        t.insert(v)
    xs = list(t)
    print('first3=', xs[:3])
    print('last2=', xs[-2:])
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 57. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Node:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None

def insert(root, val):
    if root is None:
        return Node(val)
    if val < root.val:
        root.left = insert(root.left, val)
    elif val > root.val:
        root.right = insert(root.right, val)
    return root

def inorder(t):
    if t is None:
        return
    yield from inorder(t.left)
    yield t.val
    yield from inorder(t.right)

class Tree:
    def __init__(self):
        self.root = None
    def insert(self, val):
        self.root = insert(self.root, val)
    def __iter__(self):
        return inorder(self.root)
try:
    t = Tree()
    for v in [5, 3, 8, 1, 4]:
        t.insert(v)
    print('a=', list(t))
    print('b=', list(t))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 58. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Node:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None

def insert(root, val):
    if root is None:
        return Node(val)
    if val < root.val:
        root.left = insert(root.left, val)
    elif val > root.val:
        root.right = insert(root.right, val)
    return root

def inorder(t):
    if t is None:
        return
    yield from inorder(t.left)
    yield t.val
    yield from inorder(t.right)

def preorder(t):
    if t is None:
        return
    yield t.val
    yield from preorder(t.left)
    yield from preorder(t.right)

class Tree:
    def __init__(self):
        self.root = None
    def insert(self, val):
        self.root = insert(self.root, val)
    def __iter__(self):
        return inorder(self.root)
try:
    t = Tree()
    for v in [5, 4, 3, 2, 1]:
        t.insert(v)
    print('inorder=', list(inorder(t.root)))
    print('preorder=', list(preorder(t.root)))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

Problem 59. Write the output of the final command in the following shell script.

```
$ cd; rm -rf quiz; mkdir quiz; cd quiz
$ cat > foo.py <<EOF
class Node:
    def __init__(self, val):
        self.val = val
        self.left = None
        self.right = None

def insert(root, val):
    if root is None:
        return Node(val)
    if val <= root.val:
        root.left = insert(root.left, val)
    elif val > root.val:
        root.right = insert(root.right, val)
    return root

def inorder(t):
    if t is None:
        return
    yield from inorder(t.left)
    yield t.val
    yield from inorder(t.right)

class Tree:
    def __init__(self):
        self.root = None
    def insert(self, val):
        self.root = insert(self.root, val)
    def __iter__(self):
        return inorder(self.root)
try:
    t = Tree()
    for v in [5, 3, 3]:
        t.insert(v)
    xs = list(t)
    print('list=', xs)
    print('len=', len(xs))
except TypeError:
    print('TypeError')
EOF
$ python3 foo.py
```

Fraction of LLMs with correct answer: 0 / 0 = 1.00

LLM Model Performance

